

ECOM30001/90001 Basic Econometrics

Tutorial 6: Robust SEs, Log Models & Dummy Variables

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Tutorial 6 Overview

MST Reminder

- **Date:** Wednesday, 2 September (**tomorrow**)
- **Time:** Check Canvas MST information: 2 timeslots based on your ID number
- **Content:** First topic: Estimation and Testing in the Basic Linear Regression Model

Topics in This Tutorial

- **Robust Standard Errors (Huber–White / HC1)**
- **Average Marginal Effects (AME)**
- **Reparameterization Technique**
- **Interaction Terms & Wage Differentials**
- **Delta Method**

Question (a): Wage Equation with Robust SEs

Model (1) Setup ($n = 4190$)

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + \beta_4 \text{union}_i + \beta_5 \text{metro}_i + \varepsilon_i$$

- Check R code and results, and see what difference you can tell.

Question (a): Wage Equation with Robust SEs

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Why Robust (Heteroskedasticity-Consistent) Standard Errors?

- Under heteroskedasticity ($\text{Var}(\varepsilon_i|X) \neq \sigma^2$), OLS point estimates $\hat{\beta}$ remain **unbiased and consistent**, but ordinary SEs are **invalid**.
- Robust SEs (HC1 / Huber–White) provide asymptotically valid standard errors and hypothesis tests without requiring homoskedasticity.
- **R Implementation:**
 - `cov_rob <- vcovHC(t6_eq1, type = "HC1")` (sandwich)
 - `coeftest(t6_eq1, vcov = cov_rob)` (lmtest)
- Point estimates are identical; SEs adjust for potential heteroskedasticity.

Question (b): Average Marginal Effect

Model (1) Setup ($n = 4190$)

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + \beta_4 \text{union}_i + \beta_5 \text{metro}_i + \varepsilon_i$$

- Show the Average Marginal Effect of `exper`.

Question (b): Average Marginal Effect

Partial Derivative of Model (1)

$$\frac{\partial \ln(\text{wage}_i)}{\partial \text{exper}_i} = \beta_2 + 2\beta_3 \text{exper}_i$$

Average Marginal Effect (AME):

- Because the marginal effect depends on exper_i , we evaluate it at the sample mean of experience ($\overline{\text{exper}} = 22.59$):

$$\text{AME}(\text{exper}) = \beta_2 + 2\beta_3 \overline{\text{exper}} = \beta_2 + 2(22.59)\beta_3 = \beta_2 + 45.18\beta_3$$

- Point Estimate:** Using OLS estimates from Model (1):

$$\widehat{\text{AME}}(\text{exper}) = 0.036319 + 45.18(-0.0006171) = \mathbf{0.008439} \quad (\approx 0.84\%)$$

- Interpretation:** Evaluated at the sample mean of experience (22.59 years), an additional year of experience is associated with an estimated **0.84% increase in wages, holding other factors constant** (ceteris paribus).

Question (c)-(d): Reparameterization & margins

Reparameterizing for $\beta_2 + 45.18\beta_3$

$$\begin{aligned}\ln(\text{wage}_i) &= \dots + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + \dots \\ &= \dots + (\beta_2 + 45.18\beta_3) \text{exper}_i + \beta_3 \text{exper}_i^2 - 45.18\beta_3 \text{exper}_i + \dots \\ &= \dots + (\beta_2 + 45.18\beta_3) \text{exper}_i + \beta_3 \text{exper}_i (\text{exper}_i - 45.18) + \dots\end{aligned}$$

Question (c)-(d): Reparameterization & margins

Reparameterizing for $\beta_2 + 45.18\beta_3$

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Estimation & Verification:

- **R Formula:** `lm(log(wage) ~ educ + exper + I(exper*(exper-45.18)) + union + metro, data=tut6)`
- **Coefficient on exper:** Direct estimate of $\beta_2 + 45.18\beta_3 = \mathbf{0.008439}$, robust se = **0.000512** ($t = 16.49, p < 0.001$).
- **Replication with margins:** `summary(margins(t6_eq1, variables = "exper", vcov = cov_rob))`
 $\implies$ AME = **0.0084**, se = **0.0005**, $z = 16.48$ ($p < 0.001$), matching exactly!

Question (e)-(f): Union $\times$ Metro Interaction

Model (2) with Interaction Term

$$\begin{aligned}\ln(\text{wage}_i) = & \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 \\ & + \beta_4 \text{union}_i + \beta_5 \text{metro}_i + \beta_6 (\text{union}_i \times \text{metro}_i) + \varepsilon_i\end{aligned}$$

Question (e)-(f): Union $\times$ Metro Interaction

Model (2) with Interaction Term

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Expected Log Wages for Metro-Based Workers ($\text{metro}_i = 1$):

- **Metro Union Worker ($\text{union} = 1, \text{metro} = 1$):**
 $E[\ln(\text{wage}) | \text{union}, \text{metro}] = \text{controls} + \beta_4 + \beta_5 + \beta_6$
- **Metro Non-Union Worker ($\text{union} = 0, \text{metro} = 1$):**
 $E[\ln(\text{wage}) | \text{nonunion}, \text{metro}] = \text{controls} + \beta_5$
- **Union/Non-Union Log Wage Differential in Metro Areas:**

$$\Delta_{\text{metro}} = E[\ln(\text{wage})]_{\text{union}, \text{metro}} - E[\ln(\text{wage})]_{\text{nonunion}, \text{metro}} = \beta_4 + \beta_6$$

- **Estimated Differential (from Model 2):**

$$\hat{\Delta}_{\text{metro}} = \hat{\beta}_4 + \hat{\beta}_6 = 0.1033 + (-0.0119) = \mathbf{0.0914} \quad (\approx 9.14\%)$$

Question (g): Reparameterizing for $\beta_4 + \beta_6$

Reparameterization Derivation

$$\begin{aligned}\beta_4 \text{union}_i + \beta_6(\text{union}_i \cdot \text{metro}_i) &= (\beta_4 + \beta_6)\text{union}_i + \beta_6 \text{union}_i \text{metro}_i - \beta_6 \text{union}_i \\ &= (\beta_4 + \beta_6)\text{union}_i - \beta_6 \text{union}_i(1 - \text{metro}_i) \\ &= \theta_4 \text{union}_i + \theta_6 \text{union}_i(1 - \text{metro}_i)\end{aligned}$$

Question (g): Reparameterizing for $\beta_4 + \beta_6$

Reparameterization Derivation

$$\begin{aligned}\beta_4 \text{union}_i + \beta_6 (\text{union}_i \cdot \text{metro}_i) &= (\beta_4 + \beta_6) \text{union}_i + \beta_6 \text{union}_i \text{metro}_i - \beta_6 \text{union}_i \\ &= (\beta_4 + \beta_6) \text{union}_i - \beta_6 \text{union}_i (1 - \text{metro}_i) \\ &= \theta_4 \text{union}_i + \theta_6 \text{union}_i (1 - \text{metro}_i)\end{aligned}$$

R Implementation & Results:

- **R Formula:** `lm(log(wage) ~ educ + exper + I(exper^2) + union + metro + I(union*(1-metro))), data=tut6)`
- **Regression Output (Robust SEs):**
 - Coefficient on union: $\hat{\theta}_4 = \widehat{\beta_4 + \beta_6} = \mathbf{0.09137}$
 - Robust Standard Error: $\text{se}(\hat{\theta}_4) = \text{se}(\widehat{\beta_4 + \beta_6}) = \mathbf{0.01895}$
 - $t\text{-statistic} = 4.82 \implies p = 0.00000148 < 0.001$
- **Approximate Differential:** Metro union members earn approx. **9.14%** higher log wages than non-union metro workers.

Question (h)-(i): Exact % Differential

Exact vs. Approximate Percentage Differentials

Approximate % diff = $100 \times (\hat{\beta}_4 + \hat{\beta}_6) \approx 9.14\%$.

Exact % differential: $g(\beta_4, \beta_6) = 100 \times (e^{\beta_4 + \beta_6} - 1)$

Question (h)-(i): Exact % Differential

Exact vs. Approximate Percentage Differentials

Approximate % diff = $100 \times (\hat{\beta}_4 + \hat{\beta}_6) \approx 9.14\%$.

Exact % differential: $g(\beta_4, \beta_6) = 100 \times (e^{\beta_4 + \beta_6} - 1)$

The Delta Method in R (`car::deltaMethod`):

- **R Command:** `deltaMethod(t6_eq2, "100*(exp(b4+b6)-1)",
vcov=cov_rob)`
- **Results:**
 - **Estimate:** 9.5675%
 - **Robust SE:** 2.0764%
 - **95% Confidence Interval:** [5.50%, 13.64%]
- **Conclusion:** In metro areas, union members are expected to earn an estimated **9.57% more** than their non-union counterparts, ceteris paribus.

Exam Practice Question (MCQ)

Question Setup

Consider the quadratic regression:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i \quad (3)$$

Below is a reparameterized version imposing a restriction on (3):

$$Y_i = \delta_1(1 + X_i) + \delta_2 X_i(X_i - 1) + \epsilon_i$$

What restriction is imposed on the parameters of equation (3)?

Exam Practice Question (MCQ)

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What restriction is imposed on the parameters of equation (3)?

Options to Consider:

- ① $\beta_0 + \beta_2 = \beta_1$
- ② $\beta_0 + \beta_1 + \beta_2 = 0$
- ③ $\beta_0 = \beta_1 + \beta_2$
- ④ None of the above

Exam Practice Question (MCQ Solution)

Reparameterized Equation

$$Y_i = \delta_1(1 + X_i) + \delta_2 X_i(X_i - 1) + \epsilon_i$$

Exam Practice Question (MCQ Solution)

Reparameterized Equation

$$Y_i = \delta_1(1 + X_i) + \delta_2 X_i(X_i - 1) + \epsilon_i$$

Step-by-Step Derivation:

- Expand: $Y_i = \delta_1 + \delta_1 X_i + \delta_2 X_i^2 - \delta_2 X_i + \epsilon_i = \delta_1 + (\delta_1 - \delta_2)X_i + \delta_2 X_i^2 + \epsilon_i$
- Match with $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$:

$$\beta_0 = \delta_1, \quad \beta_1 = \delta_1 - \delta_2, \quad \beta_2 = \delta_2$$

- Substitute $\delta_1 = \beta_0$ and $\delta_2 = \beta_2$ into β_1 :

$$\beta_1 = \beta_0 - \beta_2 \implies \beta_0 = \beta_1 + \beta_2$$

Exam Practice Question (MCQ Solution)

Reparameterized Equation

$$Y_i = \delta_1(1 + X_i) + \delta_2 X_i(X_i - 1) + \epsilon_i$$

Step-by-Step Derivation:

- Expand: $Y_i = \delta_1 + \delta_1 X_i + \delta_2 X_i^2 - \delta_2 X_i + \epsilon_i = \delta_1 + (\delta_1 - \delta_2)X_i + \delta_2 X_i^2 + \epsilon_i$
- Match with $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$:

$$\beta_0 = \delta_1, \quad \beta_1 = \delta_1 - \delta_2, \quad \beta_2 = \delta_2$$

- Substitute $\delta_1 = \beta_0$ and $\delta_2 = \beta_2$ into β_1 :

$$\beta_1 = \beta_0 - \beta_2 \implies \beta_0 = \beta_1 + \beta_2$$

Correct Answer

Option (3) $\beta_0 = \beta_1 + \beta_2$